

FULL TEXT LINKS

natureportfolio

Meta-Analysis Nat Hum Behav. 2024 Dec;8(12):2293-2303.

doi: 10.1038/s41562-024-02024-1. Epub 2024 Oct 28.

When combinations of humans and AI are useful: A systematic review and meta-analysis

Michelle Vaccaro ^{1 2}, Abdullah Almaatouq ¹, Thomas Malone ³

Affiliations

PMID: 39468277 PMCID: [PMC11659167](#) DOI: [10.1038/s41562-024-02024-1](#)

Abstract

Inspired by the increasing use of artificial intelligence (AI) to augment humans, researchers have studied human-AI systems involving different tasks, systems and populations. Despite such a large body of work, we lack a broad conceptual understanding of when combinations of humans and AI are better than either alone. Here we addressed this question by conducting a preregistered systematic review and meta-analysis of 106 experimental studies reporting 370 effect sizes. We searched an interdisciplinary set of databases (the Association for Computing Machinery Digital Library, the Web of Science and the Association for Information Systems eLibrary) for studies published between 1 January 2020 and 30 June 2023. Each study was required to include an original human-participants experiment that evaluated the performance of humans alone, AI alone and human-AI combinations. First, we found that, on average, human-AI combinations performed significantly worse than the best of humans or AI alone (Hedges' $g = -0.23$; 95% confidence interval, -0.39 to -0.07). Second, we found performance losses in tasks that involved making decisions and significantly greater gains in tasks that involved creating content. Finally, when humans outperformed AI alone, we found performance gains in the combination, but when AI outperformed humans alone, we found losses. Limitations of the evidence assessed here include possible publication bias and variations in the study designs analysed. Overall, these findings highlight the heterogeneity of the effects of human-AI collaboration and point to promising avenues for improving human-AI systems.

© 2024. The Author(s).

[PubMed Disclaimer](#)

Figures



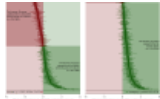


Fig. 1. Forest plots of all effect...

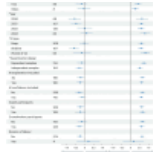


Fig. 2. Results from the three-level meta-regression...

LinkOut – more resources

Full Text Sources

[Nature Publishing Group](#)

[PubMed Central](#)